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100 MCQs — BBSHRRDB (Git/GitHub, Electronics, Python)

Section A: Version Control (Git & GitHub) — Q1 to Q20

Q1. Which command initializes a new Git repository?

- A) git start B) git init
- C) git new D) git create

Q2. Which command is used to set your global Git username?

- A) git config --global user.name "Name" B) git set username "Name"
- C) git user --global "Name" D) git config user "Name"

Q3. Which command stages a single file for commit?

- A) git stage file.txt B) git add file.txt
- C) git commit file.txt D) git push file.txt

Q4. Which command records staged changes to the repository?

- A) git save B) git push
- C) git commit D) git record

Q5. Which command uploads local commits to a remote repository?

- A) git upload B) git push
- C) git send D) git commit -m



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Q6. Which command downloads and merges changes from a remote repository?

- A) git fetch B) git download
- C) git pull D) git sync

Q7. Which command shows the current state of the working directory and staging area?

- A) git state B) git info
- C) git status D) git check

Q8. What does `git add .` do?

- A) Adds only modified files B) Adds all files in the current directory to staging
- C) Commits all files D) Deletes all staged files

Q9. Which command shows the commit history?

- A) git history B) git log
- C) git commits D) git show-all

Q10. What is GitHub primarily used for?

- A) Writing Python code B) Hosting and collaborating on Git repositories online
- C) Compiling code D) Creating databases

Q11. Which command links a local repository to a remote GitHub repository?

- A) git link remote B) git remote add origin <url>
- C) git connect <url> D) git remote set <url>

Q12. What is the difference between `git pull` and `git fetch`?

- A) They are identical B) fetch downloads changes without merging; pull downloads and merges



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C) pull only works locally D) fetch deletes the branch

Q13. Which file is commonly used to specify files that Git should ignore?

- A) .gitconfig B) .ignore
C) .gitignore D) ignore.txt

Q14. What does `git status` show for a newly created file that hasn't been staged?

- A) Modified B) Untracked
C) Committed D) Deleted

Q15. Which command creates a new branch in Git?

- A) `git branch <name>` B) `git new-branch <name>`
C) `git create-branch <name>` D) `git branch-new <name>`

Q16. Which command switches to a different branch?

- A) `git switch-to <branch>` B) `git checkout <branch>`
C) `git move <branch>` D) `git change <branch>`

Q17. What does a "commit message" represent?

- A) The file size B) A description of the changes made in that commit
C) The author's password D) The branch name

Q18. Which command shows the differences between the working directory and the last commit?

- A) `git compare` B) `git diff`
C) `git changes` D) `git delta`

Q19. What is a "remote" in Git terminology?



- A) A local backup folder B) A version of the repository hosted on a server (like GitHub)
C) A type of branch D) A commit message

Q20. Which of the following correctly clones a repository from GitHub?

- A) git copy <url> B) git clone <url>
C) git download <url> D) git fetch <url>

Section B: Basics of Electronics — Q21 to Q40

Q21. What is the unit of resistance?

- A) Volt B) Ampere
C) Ohm D) Watt

Q22. According to Ohm's Law, which formula is correct?

- A) $V = I/R$ B) $V = I \times R$
C) $I = V \times R$ D) $R = V \times I$

Q23. If $V = 10V$ and $R = 5\Omega$, what is the current I ?

- A) 0.5 A B) 2 A
C) 50 A D) 5 A

Q24. If $I = 2A$ and $R = 10\Omega$, what is the voltage V ?

- A) 5 V B) 12 V
C) 20 V D) 0.2 V

Q25. If $V = 12V$ and $I = 3A$, what is the resistance R ?



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A) $36\ \Omega$ B) $4\ \Omega$

C) $9\ \Omega$ D) $15\ \Omega$

Q26. What type of current flows in only one direction?

A) Alternating Current (AC) B) Direct Current (DC)

C) Static Current D) Reverse Current

Q27. What type of current periodically reverses direction?

A) Direct Current (DC) B) Static Electricity

C) Alternating Current (AC) D) Constant Current

Q28. In the resistor color code, what does the color "Black" represent as a digit?

A) 1 B) 0

C) 2 D) 9

Q29. In the resistor color code, what does "Brown" represent as a digit?

A) 0 B) 1

C) 2 D) 3

Q30. A resistor has bands Yellow, Violet, Red. What is its approximate value?

A) $470\ \Omega$ B) $4700\ \Omega$

C) $47000\ \Omega$ D) $47\ \Omega$

Q31. What does a Gold band typically represent on a resistor?

A) Resistance value B) Tolerance ($\pm 5\%$)

C) Power rating D) Temperature coefficient



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Q32. What does a Silver band typically represent on a resistor?

- A) Tolerance ($\pm 10\%$) B) Resistance multiplier
- C) Voltage rating D) First digit

Q33. When resistors are connected in series, the total resistance is:

- A) Less than the smallest resistor B) The sum of all resistances
- C) The average of all resistances D) Always equal to one resistor's value

Q34. When resistors are connected in parallel, the total resistance is:

- A) Greater than the largest resistor B) Always equal to the sum
- C) Less than the smallest individual resistor D) Equal to zero

Q35. Two resistors of 10Ω and 10Ω are connected in series. What is the total resistance?

- A) 5Ω B) 10Ω
- C) 20Ω D) 100Ω

Q36. Two resistors of 10Ω and 10Ω are connected in parallel. What is the total resistance?

- A) 20Ω B) 10Ω
- C) 5Ω D) 2Ω

Q37. What is the formula for total resistance of two resistors in parallel?

- A) $R = R_1 + R_2$ B) $R = R_1 \times R_2$
- C) $R = (R_1 \times R_2) / (R_1 + R_2)$ D) $R = R_1 - R_2$

Q38. Which of these is NOT a basic electronic component?

- A) Resistor B) Capacitor



C) Compiler D) Diode

Q39. Static electricity is best described as:

- A) Continuous flow of charge B) Accumulation of electric charge on a surface
C) Current that alternates direction D) Current measured in amperes only

Q40. A resistor reads Red, Red, Brown, Gold. What is its value?

- A) $22\ \Omega \pm 5\%$ B) $220\ \Omega \pm 5\%$
C) $2200\ \Omega \pm 5\%$ D) $2.2\ \Omega \pm 5\%$

Section C: Python — Q41 to Q100

Q41. Which of the following is used to define a string in Python?

- A) Single or double quotes B) Curly braces only
C) Square brackets only D) Parentheses only

Q42. What does `"Hello".upper()` return?

- A) hello B) HELLO
C) Hello D) Error

Q43. What does `len("Python")` return?

- A) 5 B) 6
C) 7 D) Error

Q44. Which method removes leading and trailing whitespace from a string?



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A) trim() B) strip()

C) clean() D) remove()

Q45. What does `"abc".replace("a", "z")` return?

A) abc B) zbc

C) abz D) Error

Q46. Which method splits a string into a list based on a delimiter?

A) split() B) divide()

C) cut() D) separate()

Q47. What is the output of `"hello"[1:4]`?

A) hel B) ell

C) ello D) hell

Q48. Which keyword is used for conditional execution in Python?

A) when B) if

C) check D) loop

Q49. Which keyword is used to check an alternate condition after an `if`?

A) elseif B) elif

C) elseiif D) elsif

Q50. What is the output of the following?

```
x = 5
if x > 10:
    print("A")
```




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```
elif x > 3:  
    print("B")  
else:  
    print("C")
```

A) A B) B

C) C D) Error

Q51. Which data type is ordered, mutable, and allows duplicates?

A) Tuple B) Set

C) List D) Dictionary

Q52. Which data type is immutable and ordered?

A) List B) Tuple

C) Dictionary D) Set

Q53. Which data type stores unique unordered elements?

A) List B) Tuple

C) Set D) String

Q54. Which data type stores key-value pairs?

A) List B) Set

C) Dictionary D) Tuple

Q55. What does `my_list.append(5)` do?

A) Removes 5 from the list B) Adds 5 to the end of the list

C) Inserts 5 at the beginning D) Replaces all elements with 5

Q56. What method is used to add an element to a set?



A) append() B) insert()

C) add() D) push()

Q57. Which method retrieves a value from a dictionary safely without raising an error if the key doesn't exist?

A) fetch() B) find()

C) get() D) retrieve()

Q58. What is the output of `len([1, 2, 3, 4])`?

A) 3 B) 4

C) 5 D) Error

Q59. Which loop is best used when the number of iterations is known?

A) while loop B) for loop

C) do-while loop D) repeat loop

Q60. What does `range(5)` generate?

A) 1,2,3,4,5 B) 0,1,2,3,4

C) 0,1,2,3,4,5 D) 1,2,3,4

Q61. What does `range(2, 10, 2)` generate?

A) 2,4,6,8 B) 2,4,6,8,10

C) 2,3,4,5,6,7,8,9 D) 2,5,8

Q62. Which loop continues executing as long as a condition is true?

A) for loop B) while loop

C) if loop D) range loop



Q63. What is the output of this code?

```
for i in range(3):  
    print(i)
```

A) 1 2 3 B) 0 1 2

C) 0 1 2 3 D) 1 2

Q64. Which keyword exits a loop prematurely?

A) stop B) exit

C) break D) end

Q65. Which keyword skips the current iteration and continues with the next?

A) skip B) continue

C) pass D) next

Q66. What will `for char in "abc": print(char)` output?

A) abc B) a b c (on separate lines)

C) Error D) "abc"

Q67. What does iterating over a dictionary using `for key in dict:` give by default?

A) Values B) Keys

C) Key-value pairs D) Index numbers

Q68. Which method returns key-value pairs together when iterating a dictionary?

A) `dict.pairs()` B) `dict.items()`

C) `dict.keys()` D) `dict.values()`



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Q69. What does `int("25")` return?

- A) "25" B) 25
C) 25.0 D) Error

Q70. What does `int("abc")` cause?

- A) Returns 0 B) Returns "abc"
C) Raises a ValueError D) Returns None

Q71. What does `list((1, 2, 3))` return?

- A) (1, 2, 3) B) [1, 2, 3]
C) {1, 2, 3} D) "1,2,3"

Q72. What does `str(100)` return?

- A) 100 B) "100"
C) 100.0 D) Error

Q73. What does `float("3.14")` return?

- A) "3.14" B) 3
C) 3.14 D) Error

Q74. What is the keyword used to define a function in Python?

- A) func B) def
C) function D) define

Q75. What does a function return if there's no explicit `return` statement?

- A) 0 B) Empty string



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C) None D) Error

Q76. What is the output of this code?

```
def add(a, b):  
    return a + b  
print(add(3, 4))
```

A) 34 B) 7

C) "3+4" D) Error

Q77. Which keyword is used to define an anonymous function?

A) func B) anon

C) lambda D) def

Q78. What is the correct lambda function for squaring a number?

A) lambda x: x*x B) lambda x = x*x

C) lambda(x): x*x D) def lambda x: x*x

Q79. Which built-in function applies a given function to every item in an iterable?

A) apply() B) map()

C) each() D) loop()

Q80. Which built-in function filters elements based on a condition?

A) select() B) filter()

C) where() D) check()

Q81. What does ***args** allow a function to accept?



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- A) A fixed number of keyword arguments B) A variable number of positional arguments
C) Only one argument D) A dictionary only

Q82. What does `**kwargs` allow a function to accept?

- A) A variable number of positional arguments B) A variable number of keyword arguments
C) Only string arguments D) A fixed list

Q83. What is a recursive function?

- A) A function that never returns B) A function that calls itself
C) A function with no parameters D) A function that runs in a loop only

Q84. What is essential for a recursive function to avoid infinite recursion?

- A) A loop B) A base case
C) A return statement only D) A global variable

Q85. What is the output of this recursive factorial call `factorial(3)` where `factorial(n) = n * factorial(n-1)` and `factorial(0)=1`?

- A) 3 B) 6
C) 9 D) 1

Q86. What is a local variable?

- A) A variable accessible throughout the entire program B) A variable defined inside a function, accessible only within it
C) A variable that never changes D) A variable defined outside any function

Q87. What is a global variable?



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A) A variable accessible only inside a function B) A variable defined outside functions, accessible throughout the program

C) A variable that cannot be modified D) A type of loop variable

Q88. Which keyword allows a function to modify a global variable?

A) global B) public

C) outer D) extern

Q89. What tool is used to install Python packages?

A) npm B) pip

C) conda only D) install.py

Q90. What command lists all installed pip packages?

A) pip show B) pip list

C) pip all D) pip packages

Q91. What is the purpose of a virtual environment (venv)?

A) To speed up code execution B) To create an isolated Python environment for project dependencies

C) To compile Python code D) To connect to the internet

Q92. Which command creates a virtual environment in Python?

A) python -m venv myenv B) python create venv

C) venv start myenv D) python venv --new

Q93. What is the purpose of a `.env` file?

A) To store executable code B) To store environment variables like API keys and secrets



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C) To store image files D) To compile Python scripts

Q94. Which package is commonly used to load variables from a `.env` file in Python?

A) `os-env` B) `python-dotenv`

C) `env-loader` D) `dotfile`

Q95. Which mode is used to open a file for reading in Python?

A) `"w"` B) `"r"`

C) `"a"` D) `"x"`

Q96. Which mode is used to append data to an existing file without overwriting it?

A) `"r"` B) `"w"`

C) `"a"` D) `"rb"`

Q97. What does `file.readlines()` return?

A) A single string of the entire file B) A list of lines in the file

C) The first line only D) The file size

Q98. What exception is commonly raised when trying to open a file that doesn't exist?

A) `ValueError` B) `FileNotFoundError`

C) `TypeError` D) `IndexError`

Q99. What is a Python "module"?

A) A built-in error type B) A file containing Python code (functions, variables, classes) that can be imported

C) A type of loop D) A function decorator



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Q100. Which keyword is used to import a custom or built-in module in Python?

- A) include B) import
C) require D) using

Answer Key

Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
1	B	26	B	51	C	76	B
2	A	27	C	52	B	77	C
3	B	28	B	53	C	78	A
4	C	29	B	54	C	79	B
5	B	30	C	55	B	80	B
6	C	31	B	56	C	81	B
7	C	32	A	57	C	82	B
8	B	33	B	58	B	83	B
9	B	34	C	59	B	84	B



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Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
10	B	35	C	60	B	85	B
11	B	36	C	61	A	86	B
12	B	37	C	62	B	87	B
13	C	38	C	63	B	88	A
14	B	39	B	64	C	89	B
15	A	40	B	65	B	90	B
16	B	41	A	66	B	91	B
17	B	42	B	67	B	92	A
18	B	43	B	68	B	93	B
19	B	44	B	69	B	94	B
20	B	45	B	70	C	95	B
21	C	46	A	71	B	96	C
22	B	47	B	72	B	97	B



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Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
23	B	48	B	73	C	98	B
24	C	49	B	74	B	99	B
25	B	50	B	75	C	100	B